

# Assignment: Normal Approximation to Binomial Distribution and Central Limit Theorem

## Instructions

- Use normal approximation with continuity correction where required.
- Express probabilities using standard normal variable  $Z$ .

## Questions

1. A coin is tossed 400 times. Find the probability that number of heads lies between 180 and 220.  
Given:  $P(0 \leq Z \leq 1.00) = 0.3413$ ,  $P(0 \leq Z \leq 2.05) = 0.4798$
2. In a factory, 10% items are defective. Out of 800 items, find the probability that defects are between 60 and 100.  
Given:  $P(0 \leq Z \leq 1.25) = 0.3944$ ,  $P(0 \leq Z \leq 2.41) = 0.4920$
3. A survey shows that 30% people prefer online shopping. In a sample of 1000 people, find probability that between 280 and 320 prefer online shopping.  
Given:  $P(0 \leq Z \leq 0.80) = 0.2881$ ,  $P(0 \leq Z \leq 1.41) = 0.4207$
4. A die is thrown 600 times. Find probability that number of times “3” appears lies between 80 and 120.  
Given:  $P(0 \leq Z \leq 0.90) = 0.3159$ ,  $P(0 \leq Z \leq 2.25) = 0.4878$
5. In a batch of 1200 bulbs, 20% are defective. Find probability that defective bulbs are between 200 and 260.  
Given:  $P(0 \leq Z \leq 1.48) = 0.4306$ ,  $P(0 \leq Z \leq 2.92) = 0.4982$
6. If  $X_1, X_2, \dots, X_{64}$  are i.i.d with mean 50 and standard deviation 8, find:

$$P(48 \leq \bar{X} \leq 52)$$

Given:  $P(0 \leq Z \leq 2.00) = 0.4772$

7. The mean weight of packets is 200 g with standard deviation 20 g. A sample of size 100 is taken. Find:

$$P(195 \leq \bar{X} \leq 205)$$

Given:  $P(0 \leq Z \leq 2.50) = 0.4938$

8. If  $X_i \sim \text{Poisson}(3)$  and  $n = 100$ , find:

$$P(280 \leq S_n \leq 320)$$

Given:  $P(0 \leq Z \leq 1.18) = 0.3810$ ,  $P(0 \leq Z \leq 1.70) = 0.4554$

9. A machine produces 5% defective items. In 1500 items, find probability that number of defectives exceeds 90.

Given:  $P(0 \leq Z \leq 1.83) = 0.4664$

10. The average marks of students is 70 with standard deviation 10. For a sample of 64 students, find:

$$P(\bar{X} > 72)$$

Given:  $P(0 \leq Z \leq 1.60) = 0.4452$